

NyayaKavach - Technical Architecture & Solution

PS-SIH26190 | Ministry of Home Affairs | Secure Document Vault

1. Executive Summary & Problem Alignment

Law enforcement agencies, courts, and forensic labs handle massive volumes of sensitive records (FIRs, Charge Sheets, Witness Statements, Forensic Reports). NyayaKavach provides an AI-driven, Zero-Trust, Blockchain-anchored digital vault providing 100% tamper proofing, automated metadata indexing, and BSA 2023 / Sec 65B IT Act legal compliance.

2. Recommended Tech Stack Selection

Frontend & UI: React.js, TailwindCSS, PDF.js, Flutter (Field Mobile Scan App)
Backend API: Node.js (Fastify), Go (API Gateway), Python FastAPI (AI Microservices)
AI & Digitization: PaddleOCR, LayoutLM, Llama-3 Legal NLP (Hindi/English PII Redaction)
Search Engine: Elasticsearch & Milvus Vector DB (Hybrid Semantic Search)
Encrypted Vault: MinIO / AWS S3 Sovereign Cloud (AES-256-GCM Envelope Encryption)
Audit Ledger: Hyperledger Fabric Consortium Blockchain (SHA-256 Merkle Hash Log)
eSign & PKI: CDAC eSign, Aadhaar eKYC, X.509 PKI Certificates

3. High-Level Architecture Topology

[Clients: Police, FSL, Prosecutor, Court] -> [Security Gateway: WAF + Keycloak IAM + ABAC Engine]
-> [Microservices: API Gateway + AI OCR + Elasticsearch + eSign Service]
-> [Data Layer: PostgreSQL Metadata + MinIO AES-256 Vault + Hyperledger Fabric Ledger]

4. Technical Workflow & Data Lifecycle

- Ingestion: Scan FIR/Forensics -> AI OCR extracts metadata & auto-redacts victim PII.
- Encryption: Client-side AES-256-GCM envelope encryption -> Save to MinIO Storage.
- Blockchain Anchoring: SHA-256 Hash + Officer ID + Timestamp -> Hyperledger Fabric Ledger.
- Zero-Trust Access: Role-Based Access Control (ABAC) with Dynamic Viewer Watermarking.
- Judicial Admissibility: Real-time SHA-256 verification + 1-Click BSA 2023 Sec 65B Certificate.

SIH Poster Content & Grid Layout

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PANEL 1: Problem Statement & Bottlenecks

- Physical & digital fragmentation across police stations, forensic labs, and courts.
- Tampering and backdating risks due to lack of cryptographic audit logs.
- PII exposure of protected witnesses and victims.
- Delays in judicial proceedings and inter-departmental transfers.

PANEL 2: Proposed Solution (NyayaKavach)

- Sovereign digital vault for criminal justice document lifecycle.
- Hyperledger Fabric blockchain ledger providing 100% tamper proof auditability.
- AI multi-lingual OCR & automated victim PII redaction.
- Fine-grained Attribute-Based Access Control (ABAC) with dynamic watermarking.

PANEL 3 & 4: Innovations & Legal Compliance

- 1-Click BSA 2023 Sec 63 / Sec 65B Electronic Evidence Certificate Generator.
- Dynamic Forensic Watermarking overlaying User ID, IP, and Timestamp.
- ICJS (Inter-operable Criminal Justice System) standardized REST/gRPC API bridge.
- Zero-Knowledge Evidence Verification for sealed court records.

PANEL 5 & 6: Measurable Impact Metrics

80% Faster Search | 100% Tamper Proof | 90% Paperless Transition

NyayaKavach Project Proposal - Smart India Hackathon (SIH) 2026